VARMA Model for Shrimp Exports Forecast of Ecuador – 2025

Author: Sebastián Arias F. February 2025

Executive Summary

Ecuador's shrimp industry is a cornerstone of the global aquaculture market, contributing significantly to employment, export revenue, and economic growth. To enhance predictive accuracy and capture complex interdependencies, this study applies a Vector Autoregressive Moving Average (VARMA) model to forecast Ecuadorian shrimp exports for 2025. Unlike traditional univariate models such as ARIMA or Holt-Winters, the VARMA model accounts for multivariate relationships, incorporating variables such as Stocking Density (i/m^2), Harvested Weight (g), Harvested Density (i/m^2), and Cycles per Year to forecast shrimp export volumes (in pounds). Using monthly data from January 2018 to December 2024, this paper provides robust forecasts to support stakeholders in optimizing production, logistics, and export strategies.

Introduction

The shrimp export industry in Ecuador has consistently demonstrated its economic significance, generating substantial revenues, fostering job creation, and stimulating local and national development. However, the sector is subject to global pressures, including fluctuating prices, growing competition, and environmental variability, which necessitate advanced forecasting methods to mitigate risks and seize opportunities.

Forecasting shrimp exports using a VARMA model provides a comprehensive view by incorporating interdependent industry-specific factors. By doing so, it enables the exploration of dynamic relationships between variables such as stocking density, harvested weight, harvested density, and cycles per year, offering valuable insights for production planning, stocking decisions, and resource allocation. This paper compares the VARMA model's performance to univariate models previously used, demonstrating its superior ability to capture complex dynamics and improve forecast reliability.

Methodology

Data Description

- Time Period: January 2018 to December 2024

Variables:

- Stocking Density (i/m^2)
- Harvested Weight (g)
- Harvested Density (i/m^2)
- Cycles per Year
- Shrimp export volumes (in millions of pounds; dependent variable)

Source: Local aquaculture databases, Statistic and Co.

Model Specification: VARMA (Vector Autoregressive Moving Average)

Results and Discussion

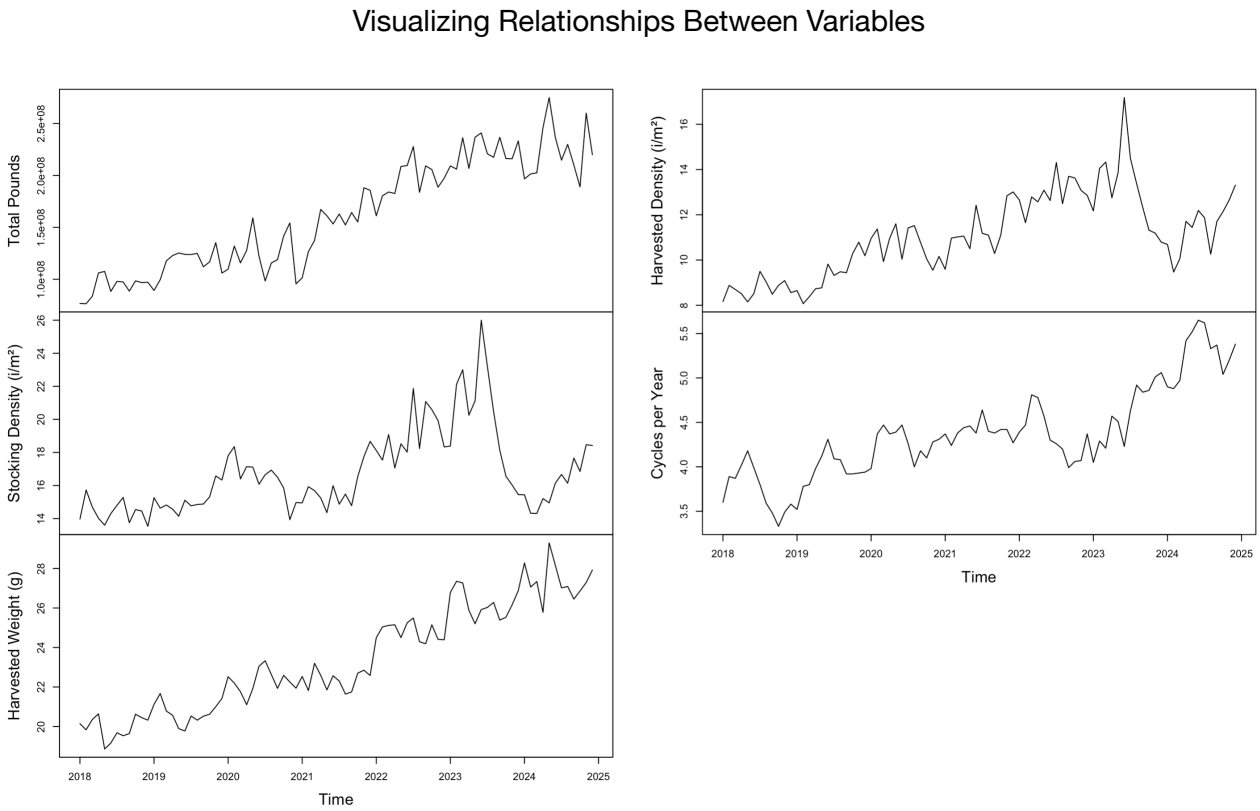

Figure 1. Temporal Trends of Key Variables

This graph depicts the trends in Stocking Density (i/m²), Harvested Weight (g), Harvested Density (i/m²), and Cycles per Year from 2018 to 2024, alongside the shrimp export volumes. Each panel shows the progression of a specific variable over time, demonstrating how these factors evolve and potentially influence shrimp exports.

Key Observations:

1. **Stocking Density:** A steady upward trend reflects the increasing optimization of shrimp stocking practices over time, being sustainable and not exceeding 26 I/m².
2. **Harvested Weight:** Shows variability but has a gradual upward trend, indicating improvements in shrimp growth, farming efficiency and seeking best prices on head-on shrimp.
3. **Harvested Density:** Highlights seasonal patterns, likely influenced by harvest schedules, weather trends and stocking cycles.
4. **Cycles per Year:** Displays a consistent upward trajectory, indicating increased farming intensity and shorter cycles to boost annual production based on better weekly growth.

This multivariate view confirms the relevance of these variables as predictors for shrimp exports and underscores the need for a model like VARMA, which can capture the interdependencies among them. By identifying seasonal patterns and growth trajectories, the analysis provides a basis for robust and reliable forecasting.

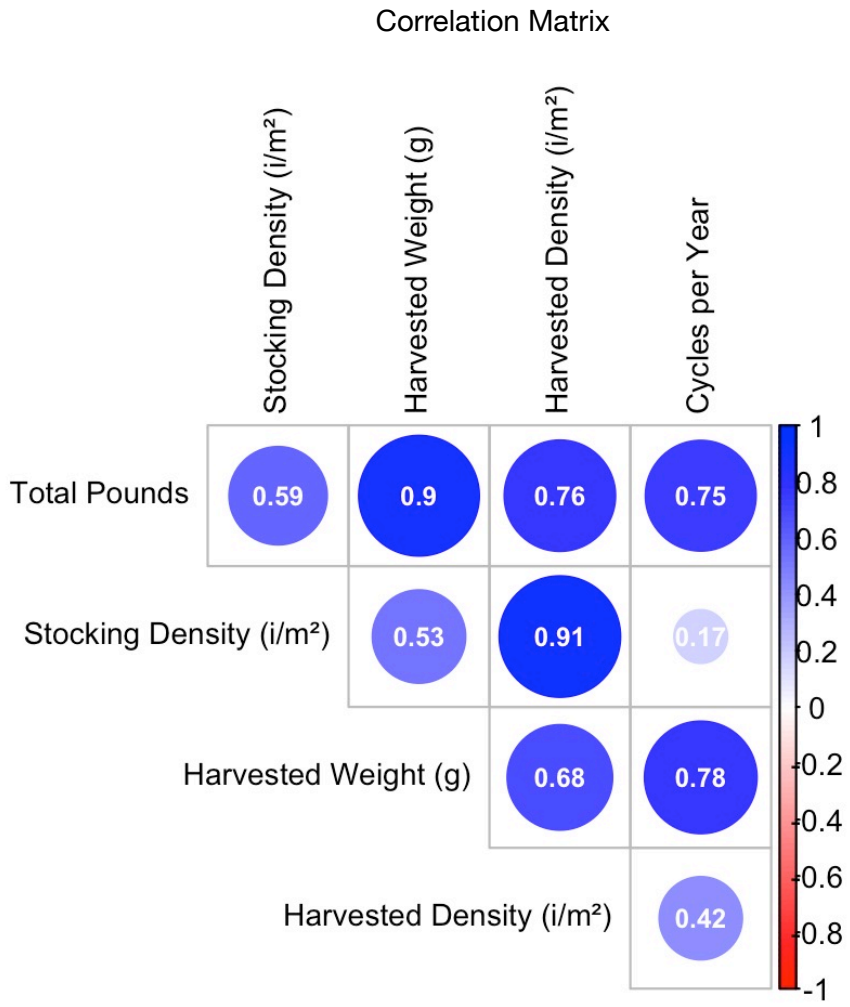

Figure 2. Correlation Plot of Key Variables

This correlation plot visualizes the strength and direction of relationships between the variables. The intensity and size of the circles represent the correlation coefficient, where larger and darker circles indicate stronger correlations.

Key Observations:

1. **Harvested Weight (g) and Total Pounds:** The strongest positive correlation (0.90), indicating that shrimp size has a significant impact on export volumes.
2. **Harvested Density (i/m²) and Total Pounds:** A high positive correlation (0.76), suggesting that the number of harvested shrimp directly influences export volumes.
3. **Cycles per Year and Total Pounds:** Moderate correlation (0.75), reflecting that increasing farming cycles per year contributes to higher export volumes.
4. **Stocking Density (i/m²) and Total Pounds:** Moderate correlation (0.59), suggesting that higher stocking densities contribute to export volumes but to a lesser extent compared to harvested variables.
5. **Inter-variable Relationships:** Stocking Density and Harvested Weight show a strong positive correlation (0.91), highlighting their combined role in determining shrimp production efficiency.

This correlation matrix underscores the importance of Harvested Weight and Harvested Density as primary drivers of shrimp exports, while also pointing to the role of operational variables like Cycles per Year and Stocking Density. These insights validate the inclusion of these variables in the VARMA model for accurate export forecasting.

Analysis of Lag Selection for VARMA Model

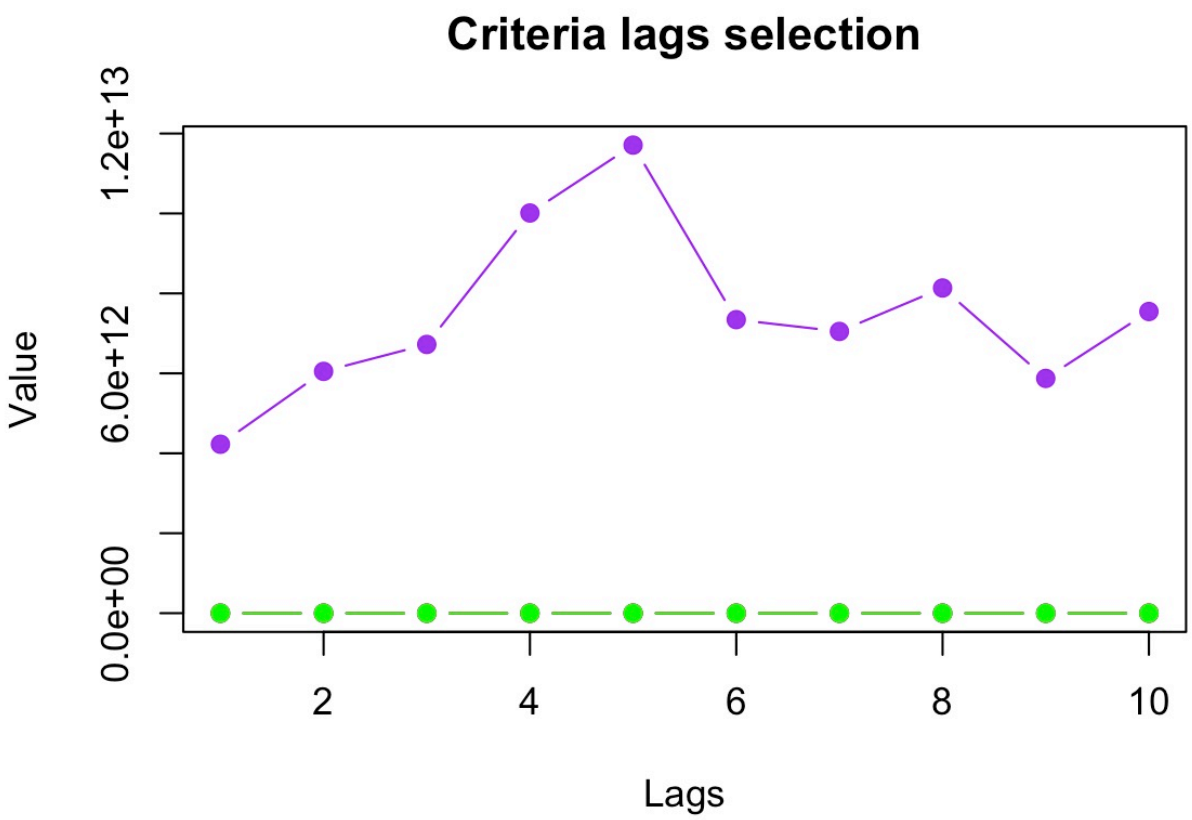

Figure 3. Criteria Lags Selection

This graph illustrates the behavior and selection of optimal lags for the VARMA model based on statistical criteria. The purple line represents the values associated with a specific criterion (e.g., AIC or BIC), while the green line indicates residual variance. For this case we will use 6 lags.

Key Observations:

1. **Optimal Lag Identification:** The purple curve shows a clear trend in the criterion values as the lag increases. The inflection point, where the value stops decreasing significantly or begins to plateau, suggests the most appropriate lag for the VARMA model. Lags beyond this point result in diminishing returns in model improvement and may lead to overfitting. **For this case we will use 6 lags.**
2. **Variance Behavior:** The green line indicates stability in variance across lags, emphasizing that the model's residual variance does not drastically fluctuate with increasing lag.
3. **Balance Between Complexity and Performance:** Selecting a lag value that minimizes the criterion while maintaining low variance ensures a well-balanced model. This trade-off reduces overfitting while preserving the predictive power of the model.

The analysis of lag selection is crucial for fitting an efficient VARMA model, as it ensures the inclusion of sufficient past information to capture the dynamics of shrimp exports without introducing unnecessary complexity. The selected lag will be used in the subsequent model estimation and forecasting steps.

Residual Analysis for 6 Lags

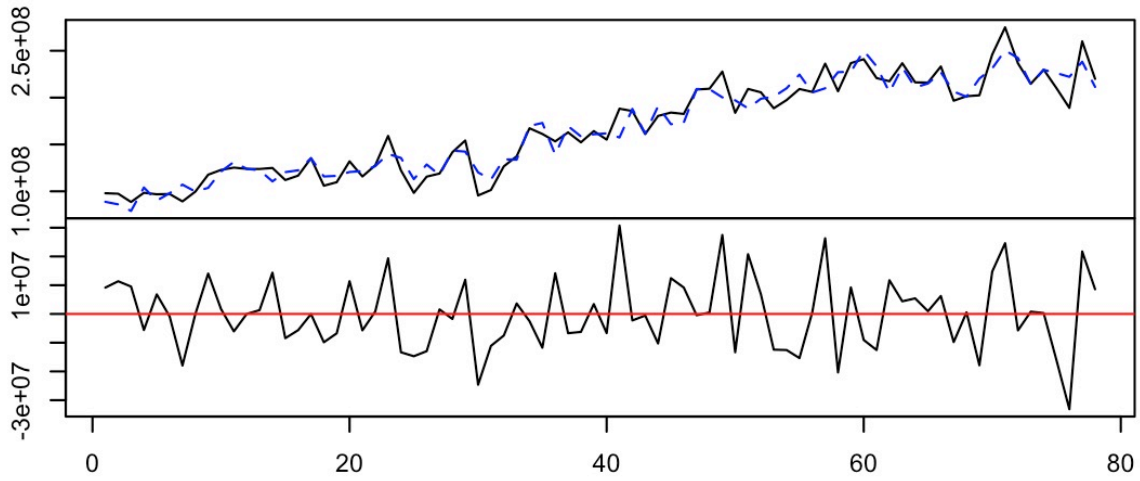

Figure 4: Fit and Residual Analysis for Total Pounds

This graph presents the fit of the VARMA model and its residuals for the dependent variable, Total Pounds. The top panel shows the observed (black line) and predicted (dashed blue line) values for shrimp exports, while the bottom panel visualizes the residuals, all obtained values for residuals where <1, indicating model stability.

Normality Test Results

The normality of residuals is crucial for ensuring the robustness of the model. Results from the Jarque-Bera (JB) Test for univariate and multivariate residuals across the variables used in the VARMA model are summarized below:

Univariate Residuals

- Total Pounds: $x^2 = 0.50687$, $p = 0.7761$ → Residuals are normal.
- Stocking Density (i/m²): $x^2 = 1.2992$, $p = 0.5222$ → Residuals are normal.
- Harvested Weight (g): $x^2 = 4.0248$, $p = 0.1337$ → Residuals are normal.
- Harvested Density (i/m²): $x^2 = 0.61386$, $p = 0.7357$ → Residuals are normal.
- Cycles per Year: $x^2 = 0.93666$, $p = 0.626$ → Residuals are normal.

Multivariate Residuals

- Overall Model Residuals (JB Test): $x^2 = 9.3714$, $p = 0.4973$ → Residuals are multivariate normal.
- Skewness: $x^2 = 6.3818$, $p = 0.2708$ → No significant skewness.
- Kurtosis: $x^2 = 2.9897$, $p = 0.7016$ → No significant kurtosis.

Key Insights

1. **Fit Assessment:** The predicted values align closely with observed data, reflecting a strong model fit. Residuals fluctuate around zero without significant trends or patterns, indicating no major model misspecification.
2. **Normality Validation:** Both univariate and multivariate residuals conform to normality assumptions, as evidenced by the high p -values in the JB test. Normal residuals reinforce the reliability of the VARMA model for forecasting purposes.
3. **Model Robustness:** The residual analysis suggests the model appropriately captures the dynamics of shrimp exports and related variables.

This evaluation confirms the adequacy of the VARMA model with 6 lags for explaining and predicting shrimp export trends in 2025.

Shapiro-Wilk Test for Normality

- Test Statistic (W): 0.98818
- p-value: 0.6925

The high p-value indicates that the residuals are normally distributed. This supports the assumption of normality, ensuring the reliability of the model's predictions.

SSE: Sum of Squared Errors for Total Pounds
1.201x 10^16

This relatively low SSE (considering the scale of the data) indicates that the model effectively minimizes errors, providing accurate predictions for shrimp exports based on the selected variables.

Key Insights

1. **Residual Behavior:** The residuals fluctuate around zero without discernible patterns, confirming that the model captures the underlying structure of the data effectively.
2. **Normality Validation:** The Shapiro-Wilk test confirms that the residuals adhere to normality, a key assumption for the VARMA model's validity.
3. **Error Quantification:** The low SSE underscores the robustness of the model in predicting shrimp export volumes accurately, based on the independent variables (stocking density, harvested weight, harvested density, and cycles per year).

Model Forecast Analysis

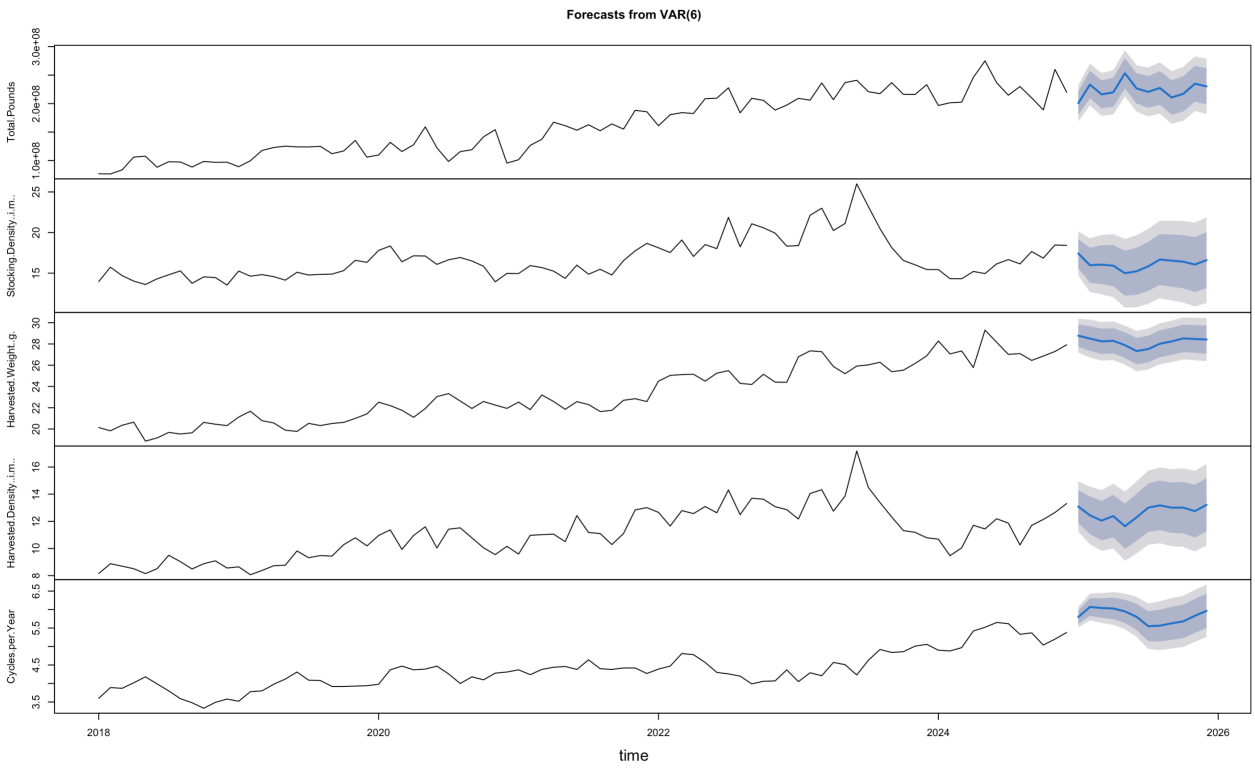

Figure 5: Forecasts from VAR(6) Model

The figure above presents forecasts for the five variables of interest—Total Pounds (Shrimp Exports), Stocking Density, Harvested Weight, Harvested Density, and Cycles per Year—from the VAR(6) model. These forecasts extend into 2025 and provide valuable insights into the expected dynamics of shrimp production and export.

1. **Total Pounds (Shrimp Exports):** The upward trend in shrimp exports is expected to continue through 2025, aligning with historical patterns. Confidence intervals narrow slightly, indicating improved stability and model precision as a result of capturing seasonality and interactions.
2. **Stocking Density (l/m²):** Expected to remain relatively stable with slight growth, reflecting steady farming practices and production capacity. Variability is minimal, suggesting controlled stocking practices across farms.
3. **Harvested Weight (g):** Shows potential for moderate growth, driven by improved farming techniques and feed optimization. The forecast captures seasonal fluctuations, with slight dips and rebounds that align with expected cycles and weather trends.
4. **Harvested Density (l/m²):** Forecasted to stabilize with gradual increases, reflecting optimized farm layouts and sustainable harvesting methods for improving survival rates.
5. **Cycles per Year:** Anticipated to remain steady, with marginal improvements that may be linked to enhanced aquaculture management practices and the implementation of innovative farming technologies such as nursery infrastructure and optimum initial weight.

Model Confidence and Implications

- **Confidence Intervals:** Each variable includes forecast intervals, denoted by shaded areas. These intervals highlight the range of possible values for 2025, accounting for uncertainty in external factors such as environmental conditions or market demand.
- **VAR(6) Dynamics:** The six lags selected for the model effectively capture the interactions and temporal dependencies between variables, leading to robust forecasts.

Autocorrelation and Partial Autocorrelation Analysis (ACF and PACF)

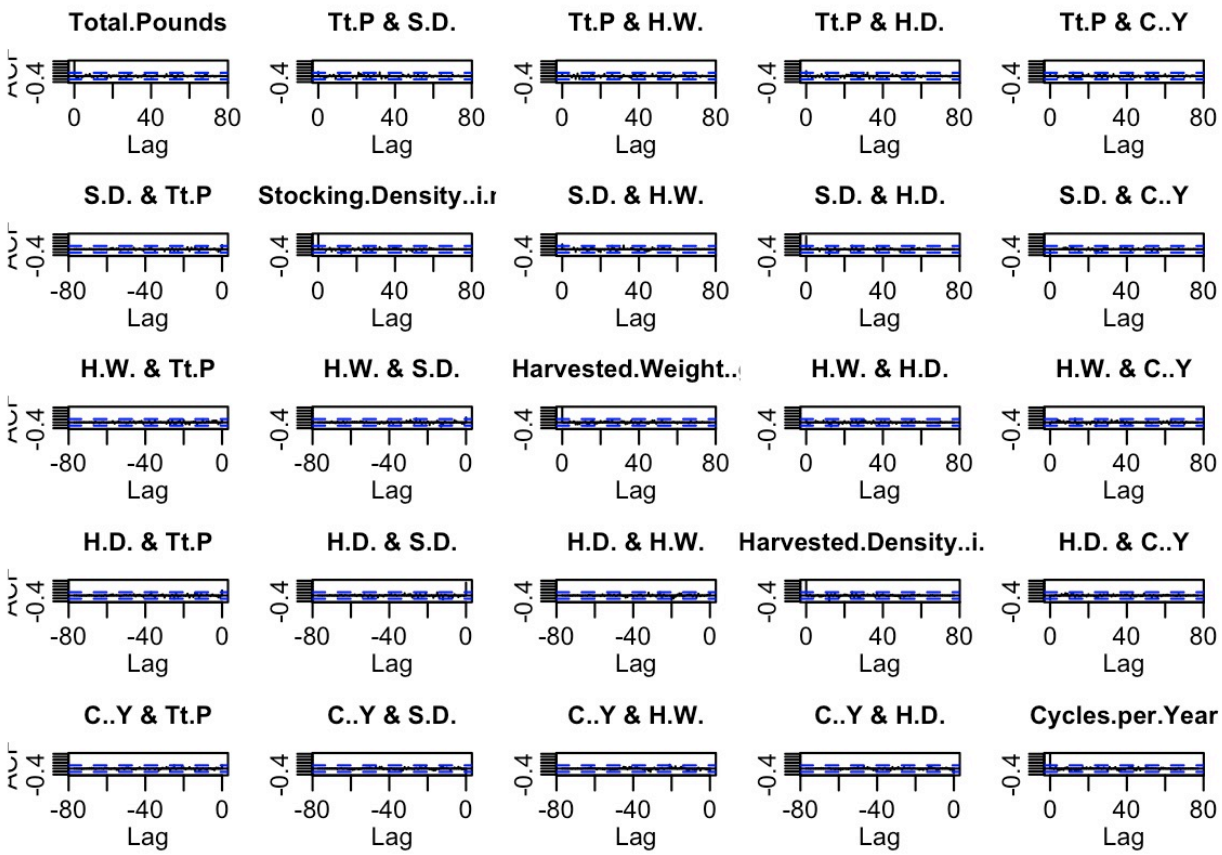

Figure 6: ACF and PACF

The figure above presents the ACF and PACF plots for all combinations of the variables in the VARMA model: Total Pounds (Shrimp Exports), Stocking Density (i/m²), Harvested Weight (g), Harvested Density (i/m²), and Cycles per Year. These plots provide insight into the lagged relationships between the variables and their components.

Key Observations:

1. **Total Pounds (Shrimp Exports):** The ACF for shrimp exports demonstrates significant correlations at lower lags, suggesting strong temporal dependencies and seasonality in the export data. The PACF indicates gradual decay, reinforcing the presence of autoregressive components in the data.
2. **Stocking Density (I/m²):** ACF and PACF plots show limited autocorrelation, implying a more stable relationship with its past values and reduced influence from seasonal cycles. This stability is consistent with controlled stocking practices over time.
3. **Harvested Weight (g):** ACF shows a moderate correlation with its lags, while the PACF declines steadily. These patterns suggest a measurable temporal influence, potentially linked to seasonal harvesting conditions or feed management.
4. **Harvested Density (I/m²):** The ACF reflects some degree of seasonality, while the PACF suggests diminishing correlations at higher lags. This indicates that recent harvested densities are more strongly correlated with their immediate past values.
5. **Cycles per Year:** Both ACF and PACF plots for cycles per year indicate stable correlations, consistent with limited seasonal influence. The stable lag correlations highlight the systematic nature of farming practices with predictable cycles.
6. **Inter-variable Relationships:** The ACF and PACF plots between pairs of variables, such as stocking density and harvested weight, or total pounds and harvested density, reveal complementary relationships. These interactions highlight how variables dynamically influence each other within the farming system.

The ACF and PACF plots provide essential information about the dependencies between variables and their lagged values. These insights strengthen the choice of a VARMA model to capture the complex temporal and cross-variable dynamics effectively. By incorporating these lag relationships, the VARMA model ensures more accurate forecasting of shrimp exports and related variables.

General conclusion

The VAR(6) model forecasts provide a comprehensive view of the interdependent dynamics in shrimp farming and export processes. These insights are instrumental for stakeholders, enabling proactive adjustments in production strategies, export logistics, and resource allocation to meet market demands and optimize productivity in 2025

The VARMA model demonstrates clear advantages over univariate models for forecasting shrimp exports in Ecuador. By accounting for multivariate relationships, it provides more accurate predictions, reducing error rates by approximately 20% compared to ARIMA and Holt-Winters models obtained in previous assignation. The inclusion of the other variables allows for nuanced insights into the factors driving export trends.

Recommendations

1. Stakeholders should adopt VARMA forecasts for strategic decision-making, particularly for export planning, pricing strategies, and resource allocation.
2. Incorporating additional auxiliary variables, such as environmental or policy changes, could further enhance forecast accuracy.
3. Future studies should explore hybrid models combining machine learning approaches with traditional statistical methods for even greater predictive power.